KONERU LAKSHMAIAH EDUCATION FOUNDATION

College of Engineering — Undergraduate Programs

ENGINEERING CAPSTONE PROJECT

RESEARCH DIARY

Capstone Project 1 (Semester VII) & Capstone Project 2 (Semester VIII)

CLUSTER A — SOFTWARE DEVELOPMENT PROJECTS

Batch	Y23 – Y27
Academic Year	
Team / Group No.	
Project Title	

This diary is the team's official evidence record. Maintain it by hand, week by week, and have your guide sign every week.

Table of Contents

Team & Guide Identification 5
Student team (3 members) 5
Guide 5
Co-guide / External mentor (if any) 5
Project identification 5
How to Use This Diary 6
Organisation of the Diary 7
Section 0 · Project Charter 8
Problem statement & motivation 9
The five objectives 10
Scope 11
Sustainability & SDG alignment (mandatory — at least one primary SDG) 12
Intended outcome vs intended impact (planned — locked at G0) 13
Resource plan 14
Section 1 · Objectives Traceability Matrix 15
Section 2 · Semester VII — Capstone Project 1 16
Phase 1 · Initiation & Charter 16
Phase 2 · Literature Survey & Positioning 19
2.1 Resource overview 19
2.2 Detailed analysis of key resources 20
2.3 Comparative analysis 22
2.4 Critical analysis 23
2.5 Gap analysis 24
2.6 Positioning / novelty statement 25
Additional space — Literature survey 26
Phase 3 · Requirements, Concept Selection & System Design 30
3.1 Requirements specification 31
3.2 Constraints & standards to comply with 32
3.3 Candidate solutions (generate multiple options) 33
3.4 Decision / trade-off matrix 34
3.5 System architecture (whole system — all five objectives) 35
3.5(a) High-level architecture diagram 36
3.5(b) Data-flow / block diagram 37
3.5(c) Component-interaction diagram 38
3.6 Detailed design of CP1 objectives (O1–O2) 39
3.7 Risk register 40
3.8 Project plan / work breakdown 41
Additional space — Requirements & design 42
Phase 4 · Implementation & Unit Testing (O1–O2) 46
4.1 Implementation progress 46
4.2 Deviations from design 47
Technical Evidence — CSE & Allied (Computing) (CP1 · objectives O1–O2) 48
Requirements & Architecture (ISO/IEC/IEEE 29148 · 42010) 48
Architecture / component & data-flow diagram 49
Data & Model Evidence (AI&DS / data projects) 50
Build & Version Control 51
Security & Quality (OWASP / ISO 27001 · ISO/IEC 25010) 52
4.3 Unit test plan 53
4.4 Unit test execution 54
4.5 Issue tracker 55
Additional space — Implementation & testing (CP1) 56
Phase 5 · Consolidation, Compliance & PDR 61
5.1 Initial integration (O1 + O2) 61
5.2 Standards identification & compliance checklist 62
CP1 Reflection & Handoff to CP2 66
Section 3 · Semester VIII — Capstone Project 2 68
Phase 1 · Re-baseline, Design Refinement & CDR 68
1.1 Literature update (new work since CP1) 69
1.2 Design change log (what changed from CP1 and why) 70
1.3 Detailed design of remaining objectives (O3–O5) 71

1.4 Refined / updated architecture diagram (if changed) 72
1.5 Updated risk register 73
1.6 Test strategy (unit → integration → system) 74
Phase 2 · Implementation of Remaining Objectives (O3–O5) 78
2.1 Implementation progress 78
2.2 Deviations from design 79
Technical Evidence — IT & Allied (Computing) (CP2 · objectives O3–O5) 80
Requirements & Architecture (ISO/IEC/IEEE 29148 · 42010) 80
Architecture / component & data-flow diagram 81
Data & Model Evidence (AI&DS / data projects) 82
Build & Version Control 83
Security & Quality (OWASP / ISO 27001 · ISO/IEC 25010) 84
2.3 Unit test execution (O3–O5) 85
2.4 Issue tracker 86
Additional space — Implementation & testing (CP2) 87
Phase 3 · Integration & System-Level Testing 93
3.1 System integration (all objectives together) 93
3.2 System-level test plan & execution 94
3.3 Experiment / investigation record (generate & analyse data) 95
3.4 Standards-compliance validation 96
Additional space — Integration & system testing 97
Phase 4 · Results & Impact Evaluation 101
4.1 Outcome — planned vs achieved (per objective) 101
4.2 Results analysis & benchmarking 102
4.3 Impact — intended vs created 103
Phase 5 · Dissemination, Documentation & Final Defense 106
5.1 Dissemination 106
5.2 Individual contribution log (guide-verified) 106
5.3 Final ethics / data / IP confirmation 107
CP2 Reflection & Future Work 110
Section 4 · Standards, Outcome Mapping & Evidence Index 112
1.1 Programme Outcome (PO) mapping 112
1.2 Programme-Specific Outcome (PSO) mapping — this cluster 113
1.3 Standards register (project-wide) 114
1.4 Evidence index 115
Annex · Cluster A — Programme-Specific Notes 116
Indicative standards & codes for this cluster 116

Team & Guide Identification

Student team (3 members)

Sl	Student Name	Student ID	Program	Role
1				
2				
3				

Guide

Guide Employee ID	
Guide Name	
Department	
Signature & Date	

Co-guide / External mentor (if any)

Name & Affiliation	
Signature & Date	

Project identification

Technology domain	
Project title	
Cluster	Cluster A — All Software Projects

How to Use This Diary

- Print and bind this diary as one hard copy per team of three. It travels with the team through both semesters.
- Fill it in as the work happens — not at the end. It is a running record, not a report written afterwards.
- Every week, complete a Weekly Progress Log: what was done, who did it, challenges, evidence produced, and the guide's comments. The guide signs each week — that signature is the proof that the team met the guide.
- The diary records the process and evidence pointers (commit IDs, drawing numbers, test-case IDs, lab-notebook pages), not the full code or product. Keep the actual artefacts in your repository / lab and reference them here.
- Objectives are the unit of everything. Give each objective an ID (O1–O5) and trace it from requirement → design → build → test → result in the Traceability Matrix.
- Be honest about outcomes and impact. Record the outcome you planned versus what you achieved, and the impact you intended versus what you actually created — including the gap and the pathway to close it. Honest gap analysis scores better than an unsupported claim.
- Where a section runs out of space, use the 'Additional space' pages provided (or attach a numbered annexure sheet) rather than cramping your writing.
- Reviews are gates. You must clear Review 1 and the Preliminary Design Review in CP1, and the Critical Design Review and Final Defense in CP2.

Organisation of the Diary

- Section 0 — Project Charter (locked at Gate G0, Week 2)
- Section 1 — Objectives Traceability Matrix (living, both semesters)
- Section 2 — Semester VII · Capstone Project 1 (Weeks 1–15, Phases 1–5, Reviews R1 & PDR)
- Section 3 — Semester VIII · Capstone Project 2 (Weeks 1–15, Phases 1–5, CDR & Final Defense)
- Section 4 — Standards Register, CO–PO–PSO Mapping & Evidence Index
- Annex — Program-specific outcomes (PSOs) and standards for this cluster

Section 0 · Project Charter

Complete during Weeks 1–2 and get it approved (signed) at Gate G0. Once signed, the 'planned' fields are locked — record later changes in the CP2 change log instead.

Problem statement & motivation

What real problem does this project address, and why does it matter?

The five objectives

ID	Objective	Measurable success criterion	Owner(s)	Sem	Deliverable
O1					
O2					
O3					
O4					
O5					

Suggested split: O1–O2 in CP1 (Sem VII); O3–O5 + integration in CP2 (Sem VIII). Adjust to the size of each objective.

Scope

In scope	Out of scope

Sustainability & SDG alignment (mandatory — at least one primary SDG)

Primary SDG & target	*e.g., SDG 6.4 — water-use efficiency*
Intended beneficiary	
Justification of linkage	
Secondary SDG(s) (optional)	

Intended outcome vs intended impact (planned — locked at G0)

Intended OUTCOME (technical result to be delivered)	Intended IMPACT (change aimed for + baseline)

Resource plan

Resource (SW/HW/data/lab)	Purpose	Source / provider	Availability status

GATE G0 — CHARTER APPROVAL
Guide approval to proceed (Signature & Date): *Verified at Review 1*

Section 1 · Objectives Traceability Matrix

The backbone of the diary. Update it every week. Every objective must trace from requirement to evidence, with a status and a guide sign-off. Status: Not started / In progress / Achieved / Partially achieved / Dropped (with justification).

ID	Success criterion	Req. ref	Design ref	Build ref	Test ref	Status	Guide sign
O1							
O2							
O3							
O4							
O5							

Section 2 · Semester VII — Capstone Project 1

Objective of CP1: start the project from scratch, produce the full design of all five objectives, and build and unit-test the first objectives (O1–O2). Two panel reviews: Review 1 (Week 6, feasibility) and the Preliminary Design Review (Week 15).

Phase 1 · Initiation & Charter	
Weeks	Weeks 1–2
Gate / Panel review	Gate G0 — Charter approval
Accreditation coverage	PO1, PO6, PO11 · ABET SO2, SO4

- Deliverable this phase: the completed, guide-signed Project Charter (Section 0) and an opened Traceability Matrix (Section 1).

SEMESTER VII · CP1 · WEEKLY PROGRESS LOG	
Week No.	01
Phase	Phase 1 — Initiation & Charter
Dates (from – to)	
Planned agenda / focus for the week	

Member-wise work done this week (who did what)

Member (Name)	Student ID	Tasks undertaken this week	Prog.%
1.			
2.			
3.			

Actionable items completed

Challenges faced & how they were addressed

Evidence produced this week (repo/commit, drawing/schematic no., test-case ID, lab-notebook page, dataset, photo…)

GUIDE FEEDBACK / COMMENTS (record of the weekly meeting)

Action plan for next week

Task	Expected outcome	Owner	Due date

Guide Signature & Date	Team Lead Signature & Date

SEMESTER VII · CP1 · WEEKLY PROGRESS LOG	
Week No.	02
Phase	Phase 1 — Initiation & Charter
Dates (from – to)	
Planned agenda / focus for the week	

Member-wise work done this week (who did what)

Member (Name)	Student ID	Tasks undertaken this week	Prog.%
1.			
2.			
3.			

Actionable items completed

Challenges faced & how they were addressed

Evidence produced this week (repo/commit, drawing/schematic no., test-case ID, lab-notebook page, dataset, photo…)

GUIDE FEEDBACK / COMMENTS (record of the weekly meeting)

Action plan for next week

Task	Expected outcome	Owner	Due date

Guide Signature & Date	Team Lead Signature & Date

Phase 2 · Literature Survey & Positioning	
Weeks	Weeks 3–5
Gate / Panel review	Guide sign-off on survey
Accreditation coverage	PO4, PO11 · ABET SO6, SO7 (WK8)

2.1 Resource overview

Sl	Source title / citation	Year	Type / indexer (SCI/Scopus/IEEE/…)

2.2 Detailed analysis of key resources

One block per key resource. Duplicate a block (or use the Additional-space page) if you have more.

Resource 01

Title / Author(s) / DOI	
Key focus & methodology	
Tools / technologies	
Outcomes & limitations	
Relevance to our project & learning	

Resource 02

Title / Author(s) / DOI	
Key focus & methodology	
Tools / technologies	
Outcomes & limitations	
Relevance to our project & learning	

Resource 03

Title / Author(s) / DOI	
Key focus & methodology	
Tools / technologies	
Outcomes & limitations	
Relevance to our project & learning	

Resource 04

Title / Author(s) / DOI	
Key focus & methodology	
Tools / technologies	
Outcomes & limitations	
Relevance to our project & learning	

2.3 Comparative analysis

Article	Source	Key findings	Relevance to our design

2.4 Critical analysis

Resource	Strengths	Weaknesses	Applicability

2.5 Gap analysis

Gap / problem area	Insight from literature	How our project addresses it

2.6 Positioning / novelty statement

Where our project sits relative to existing work.

Guide sign-off on literature survey (Signature & Date)	

Additional space — Literature survey

Use this page to continue any table above or to record extra findings for this section. Note which item/row this continues.

SEMESTER VII · CP1 · WEEKLY PROGRESS LOG	
Week No.	03
Phase	Phase 2 — Literature Survey & Positioning
Dates (from – to)	
Planned agenda / focus for the week	

Member-wise work done this week (who did what)

Member (Name)	Student ID	Tasks undertaken this week	Prog.%
1.			
2.			
3.			

Actionable items completed

Challenges faced & how they were addressed

Evidence produced this week (repo/commit, drawing/schematic no., test-case ID, lab-notebook page, dataset, photo…)

GUIDE FEEDBACK / COMMENTS (record of the weekly meeting)

Action plan for next week

Task	Expected outcome	Owner	Due date

Guide Signature & Date	Team Lead Signature & Date

SEMESTER VII · CP1 · WEEKLY PROGRESS LOG	
Week No.	04
Phase	Phase 2 — Literature Survey & Positioning
Dates (from – to)	
Planned agenda / focus for the week	

Member-wise work done this week (who did what)

Member (Name)	Student ID	Tasks undertaken this week	Prog.%
1.			
2.			
3.			

Actionable items completed

Challenges faced & how they were addressed

Evidence produced this week (repo/commit, drawing/schematic no., test-case ID, lab-notebook page, dataset, photo…)

GUIDE FEEDBACK / COMMENTS (record of the weekly meeting)

Action plan for next week

Task	Expected outcome	Owner	Due date

Guide Signature & Date	Team Lead Signature & Date

SEMESTER VII · CP1 · WEEKLY PROGRESS LOG	
Week No.	05
Phase	Phase 2 — Literature Survey & Positioning
Dates (from – to)	
Planned agenda / focus for the week	

Member-wise work done this week (who did what)

Member (Name)	Student ID	Tasks undertaken this week	Prog.%
1.			
2.			
3.			

Actionable items completed

Challenges faced & how they were addressed

Evidence produced this week (repo/commit, drawing/schematic no., test-case ID, lab-notebook page, dataset, photo…)

GUIDE FEEDBACK / COMMENTS (record of the weekly meeting)

Action plan for next week

Task	Expected outcome	Owner	Due date

Guide Signature & Date	Team Lead Signature & Date

Phase 3 · Requirements, Concept Selection & System Design	
Weeks	Weeks 6–8
Gate / Panel review	Review 1 @ Week 6 (feasibility gate)
Accreditation coverage	PO2, PO3, PO5 · ABET SO1, SO2

3.1 Requirements specification

Req. ID	Type (Functional / Non-functional)	Requirement	Traces to objective

3.2 Constraints & standards to comply with

Constraint / standard	Type	How it shapes the design

3.3 Candidate solutions (generate multiple options)

Concept	Description	Pros	Cons

3.4 Decision / trade-off matrix

Criterion	Weight	Concept A	Concept B	Concept C

Selected concept & justification

3.5 System architecture (whole system — all five objectives)

Provide the three views below; each has a full page. Draw to scale and label clearly.

3.5(a) High-level architecture diagram

(Draw or paste the diagram here — include a title, labels and a legend.)

3.5(b) Data-flow / block diagram

(Draw or paste the diagram here — include a title, labels and a legend.)

3.5(c) Component-interaction diagram

(Draw or paste the diagram here — include a title, labels and a legend.)

3.6 Detailed design of CP1 objectives (O1–O2)

Component	Description	Inputs	Outputs	Dependencies

3.7 Risk register

Risk ID	Description	Likelihood	Impact	Mitigation	Owner

3.8 Project plan / work breakdown

Phase / Task	Owner(s)	Start	End	Dependencies

Additional space — Requirements & design

Use this page to continue any table above or to record extra findings for this section. Note which item/row this continues.

SEMESTER VII · CP1 · WEEKLY PROGRESS LOG	
Week No.	06
Phase	Phase 3 — Requirements, Concept Selection & Design
Dates (from – to)	
Planned agenda / focus for the week	

Member-wise work done this week (who did what)

Member (Name)	Student ID	Tasks undertaken this week	Prog.%
1.			
2.			
3.			

Actionable items completed

Challenges faced & how they were addressed

Evidence produced this week (repo/commit, drawing/schematic no., test-case ID, lab-notebook page, dataset, photo…)

GUIDE FEEDBACK / COMMENTS (record of the weekly meeting)

Action plan for next week

Task	Expected outcome	Owner	Due date

Guide Signature & Date	Team Lead Signature & Date

SEMESTER VII · CP1 · WEEKLY PROGRESS LOG	
Week No.	07
Phase	Phase 3 — Requirements, Concept Selection & Design
Dates (from – to)	
Planned agenda / focus for the week	

Member-wise work done this week (who did what)

Member (Name)	Student ID	Tasks undertaken this week	Prog.%
1.			
2.			
3.			

Actionable items completed

Challenges faced & how they were addressed

Evidence produced this week (repo/commit, drawing/schematic no., test-case ID, lab-notebook page, dataset, photo…)

GUIDE FEEDBACK / COMMENTS (record of the weekly meeting)

Action plan for next week

Task	Expected outcome	Owner	Due date

Guide Signature & Date	Team Lead Signature & Date

SEMESTER VII · CP1 · WEEKLY PROGRESS LOG	
Week No.	08
Phase	Phase 3 — Requirements, Concept Selection & Design
Dates (from – to)	
Planned agenda / focus for the week	

Member-wise work done this week (who did what)

Member (Name)	Student ID	Tasks undertaken this week	Prog.%
1.			
2.			
3.			

Actionable items completed

Challenges faced & how they were addressed

Evidence produced this week (repo/commit, drawing/schematic no., test-case ID, lab-notebook page, dataset, photo…)

GUIDE FEEDBACK / COMMENTS (record of the weekly meeting)

Action plan for next week

Task	Expected outcome	Owner	Due date

Guide Signature & Date	Team Lead Signature & Date

Phase 4 · Implementation & Unit Testing (O1–O2)	
Weeks	Weeks 9–12
Gate / Panel review	Continuous guide assessment
Accreditation coverage	PO3, PO5 · ABET SO2, SO6

4.1 Implementation progress

Task	Owner(s)	Start	End	% done	Dependencies

4.2 Deviations from design

Objective / module	Deviation from design	Reason	Impact

Technical Evidence — CSE & Allied (Computing) (CP1 · objectives O1–O2)

Record the discipline-specific engineering evidence for the objectives built in this phase. Each item below has its own page(s); duplicate any table if you need more rows.

Requirements & Architecture (ISO/IEC/IEEE 29148 · 42010)

Req. ID	Type (F/NF)	Requirement / quality attribute	Priority	Traces to objective

Architecture / component & data-flow diagram

Note the ISO/IEC 25010 quality attributes (performance, security, reliability, maintainability…) the design addresses.

(Draw or paste the diagram here — include a title, labels and a legend.)

Data & Model Evidence (AI&DS / data projects)

Dataset(s)

Dataset / source	Size & split	Licence & provenance	Pre-processing / issues

Experiments

Experiment / run ID	Config / model	Metric	Baseline	Result vs target

Bias / fairness & data-privacy check (consent, anonymisation, DPDP/GDPR relevance)

Build & Version Control

Module / component	Owner	Repository / commit ref	CI-CD / build status

Security & Quality (OWASP / ISO 27001 · ISO/IEC 25010)

Control / quality attribute	Applicable? (Y/N)	How addressed / evidence

4.3 Unit test plan

Test-case ID	Description	Objective / module	Test type

4.4 Unit test execution

Test-case ID	Input	Expected output	Actual output	Pass/Fail

4.5 Issue tracker

Issue no.	Description	Impact	Owner	Status	Guide sign

Additional space — Implementation & testing (CP1)

Use this page to continue any table above or to record extra findings for this section. Note which item/row this continues.

SEMESTER VII · CP1 · WEEKLY PROGRESS LOG	
Week No.	09
Phase	Phase 4 — Implementation & Unit Testing (O1–O2)
Dates (from – to)	
Planned agenda / focus for the week	

Member-wise work done this week (who did what)

Member (Name)	Student ID	Tasks undertaken this week	Prog.%
1.			
2.			
3.			

Actionable items completed

Challenges faced & how they were addressed

Evidence produced this week (repo/commit, drawing/schematic no., test-case ID, lab-notebook page, dataset, photo…)

GUIDE FEEDBACK / COMMENTS (record of the weekly meeting)

Action plan for next week

Task	Expected outcome	Owner	Due date

Guide Signature & Date	Team Lead Signature & Date

SEMESTER VII · CP1 · WEEKLY PROGRESS LOG	
Week No.	10
Phase	Phase 4 — Implementation & Unit Testing (O1–O2)
Dates (from – to)	
Planned agenda / focus for the week	

Member-wise work done this week (who did what)

Member (Name)	Student ID	Tasks undertaken this week	Prog.%
1.			
2.			
3.			

Actionable items completed

Challenges faced & how they were addressed

Evidence produced this week (repo/commit, drawing/schematic no., test-case ID, lab-notebook page, dataset, photo…)

GUIDE FEEDBACK / COMMENTS (record of the weekly meeting)

Action plan for next week

Task	Expected outcome	Owner	Due date

Guide Signature & Date	Team Lead Signature & Date

SEMESTER VII · CP1 · WEEKLY PROGRESS LOG	
Week No.	11
Phase	Phase 4 — Implementation & Unit Testing (O1–O2)
Dates (from – to)	
Planned agenda / focus for the week	

Member-wise work done this week (who did what)

Member (Name)	Student ID	Tasks undertaken this week	Prog.%
1.			
2.			
3.			

Actionable items completed

Challenges faced & how they were addressed

Evidence produced this week (repo/commit, drawing/schematic no., test-case ID, lab-notebook page, dataset, photo…)

GUIDE FEEDBACK / COMMENTS (record of the weekly meeting)

Action plan for next week

Task	Expected outcome	Owner	Due date

Guide Signature & Date	Team Lead Signature & Date

SEMESTER VII · CP1 · WEEKLY PROGRESS LOG	
Week No.	12
Phase	Phase 4 — Implementation & Unit Testing (O1–O2)
Dates (from – to)	
Planned agenda / focus for the week	

Member-wise work done this week (who did what)

Member (Name)	Student ID	Tasks undertaken this week	Prog.%
1.			
2.			
3.			

Actionable items completed

Challenges faced & how they were addressed

Evidence produced this week (repo/commit, drawing/schematic no., test-case ID, lab-notebook page, dataset, photo…)

GUIDE FEEDBACK / COMMENTS (record of the weekly meeting)

Action plan for next week

Task	Expected outcome	Owner	Due date

Guide Signature & Date	Team Lead Signature & Date

Phase 5 · Consolidation, Compliance & PDR	
Weeks	Weeks 13–15
Gate / Panel review	Preliminary Design Review @ Week 15
Accreditation coverage	PO7, PO9, PO10 · ABET SO3, SO4

5.1 Initial integration (O1 + O2)

Subsystems involved	Integration objective	Result	Issues identified

5.2 Standards identification & compliance checklist

Standard / protocol	Relevance to project	Compliance step / evidence

5.3 Ethics, data-privacy, IP & academic-integrity declaration

- We have followed the applicable ethical, data-privacy and safety norms for this project.
- All third-party code, data, designs and content are appropriately licensed and cited.
- The work is our own; any AI assistance used is disclosed below with how it was used.
- We have run a similarity/plagiarism check on written material and retained the report.

Disclosure of AI tools used & how (models, purpose, extent)

Member 1 — Sign & Date	Member 2 — Sign & Date	Member 3 — Sign & Date

SEMESTER VII · CP1 · WEEKLY PROGRESS LOG	
Week No.	13
Phase	Phase 5 — Consolidation, Compliance & PDR
Dates (from – to)	
Planned agenda / focus for the week	

Member-wise work done this week (who did what)

Member (Name)	Student ID	Tasks undertaken this week	Prog.%
1.			
2.			
3.			

Actionable items completed

Challenges faced & how they were addressed

Evidence produced this week (repo/commit, drawing/schematic no., test-case ID, lab-notebook page, dataset, photo…)

GUIDE FEEDBACK / COMMENTS (record of the weekly meeting)

Action plan for next week

Task	Expected outcome	Owner	Due date

Guide Signature & Date	Team Lead Signature & Date

SEMESTER VII · CP1 · WEEKLY PROGRESS LOG	
Week No.	14
Phase	Phase 5 — Consolidation, Compliance & PDR
Dates (from – to)	
Planned agenda / focus for the week	

Member-wise work done this week (who did what)

Member (Name)	Student ID	Tasks undertaken this week	Prog.%
1.			
2.			
3.			

Actionable items completed

Challenges faced & how they were addressed

Evidence produced this week (repo/commit, drawing/schematic no., test-case ID, lab-notebook page, dataset, photo…)

GUIDE FEEDBACK / COMMENTS (record of the weekly meeting)

Action plan for next week

Task	Expected outcome	Owner	Due date

Guide Signature & Date	Team Lead Signature & Date

SEMESTER VII · CP1 · WEEKLY PROGRESS LOG	
Week No.	15
Phase	Phase 5 — Consolidation, Compliance & PDR
Dates (from – to)	
Planned agenda / focus for the week	

Member-wise work done this week (who did what)

Member (Name)	Student ID	Tasks undertaken this week	Prog.%
1.			
2.			
3.			

Actionable items completed

Challenges faced & how they were addressed

Evidence produced this week (repo/commit, drawing/schematic no., test-case ID, lab-notebook page, dataset, photo…)

GUIDE FEEDBACK / COMMENTS (record of the weekly meeting)

Action plan for next week

Task	Expected outcome	Owner	Due date

Guide Signature & Date	Team Lead Signature & Date

Capstone Project 1 — Guide Evaluation Rubric

Continuous (guide) + review (panel) assessment. Guide circles the band and enters a score out of 10 for each criterion; the weighted total gives the phase grade.

Performance band	What it means (applies to every criterion)
Exemplary (9–10)	Complete, rigorous and well-evidenced; independent, professional-quality work that exceeds expectations.
Proficient (7–8)	Meets all expectations with sound method and adequate evidence; minor gaps only.
Developing (5–6)	Partially meets expectations; method or evidence incomplete; needs guidance to progress.
Inadequate (<5)	Missing, superficial or unsupported; does not meet the expected standard.

Criterion	Wt %	Evidence to look for	Band / Score /10
Problem formulation & scoping	10	Clear, feasible, well-scoped problem with measurable objectives	
Literature survey & positioning	15	Recent, credible sources; genuine gap; novelty statement	
Requirements & design (multiple solutions + trade-off)	20	Traceable requirements; ≥2 concepts; justified decision matrix; whole-system design	
Implementation & unit testing (O1–O2)	20	Two objectives built and unit-tested with evidence and deviation notes	
Standards, ethics & sustainability/SDG	10	Relevant standards applied; ethics/AI-use declared; SDG linkage justified	
Project management & teamwork	10	Realistic plan, risk register, balanced individual contribution	
Documentation & communication	15	Complete diary, guide sign-offs, review performance	
Total	**100**		

Overall guide comments & areas for improvement

Guide Signature & Date	

CP1 Reflection & Handoff to CP2

What went well, what was hard, and key lessons from CP1

Handoff to CP2 — remaining objectives (O3–O5), revised plan and design changes to carry forward

Section 3 · Semester VIII — Capstone Project 2

Objective of CP2: refine and freeze the design, build the remaining objectives (O3–O5), integrate the full system, validate it, evidence the real outcome and impact, and defend it. Two panel reviews: Critical Design Review (Week 3) and Final Defense (Week 15).

Phase 1 · Re-baseline, Design Refinement & CDR	
Weeks	Weeks 1–3
Gate / Panel review	Critical Design Review @ Week 3 (design freeze)
Accreditation coverage	PO3, PO4 · ABET SO6

1.1 Literature update (new work since CP1)

Source	Year	New insight	Effect on our design

1.2 Design change log (what changed from CP1 and why)

Change ID	What changed	Reason	Impact on objectives/plan

1.3 Detailed design of remaining objectives (O3–O5)

Component	Description	Inputs	Outputs	Dependencies

1.4 Refined / updated architecture diagram (if changed)

Mark what changed from the CP1 design.

(Draw or paste the diagram here — include a title, labels and a legend.)

1.5 Updated risk register

Risk ID	Description	Likelihood	Impact	Mitigation	Owner

1.6 Test strategy (unit → integration → system)

Test strategy for the remaining build and full-system validation

SEMESTER VIII · CP2 · WEEKLY PROGRESS LOG	
Week No.	01
Phase	Phase 1 — Re-baseline, Design Refinement & CDR
Dates (from – to)	
Planned agenda / focus for the week	

Member-wise work done this week (who did what)

Member (Name)	Student ID	Tasks undertaken this week	Prog.%
1.			
2.			
3.			

Actionable items completed

Challenges faced & how they were addressed

Evidence produced this week (repo/commit, drawing/schematic no., test-case ID, lab-notebook page, dataset, photo…)

GUIDE FEEDBACK / COMMENTS (record of the weekly meeting)

Action plan for next week

Task	Expected outcome	Owner	Due date

Guide Signature & Date	Team Lead Signature & Date

SEMESTER VIII · CP2 · WEEKLY PROGRESS LOG	
Week No.	02
Phase	Phase 1 — Re-baseline, Design Refinement & CDR
Dates (from – to)	
Planned agenda / focus for the week	

Member-wise work done this week (who did what)

Member (Name)	Student ID	Tasks undertaken this week	Prog.%
1.			
2.			
3.			

Actionable items completed

Challenges faced & how they were addressed

Evidence produced this week (repo/commit, drawing/schematic no., test-case ID, lab-notebook page, dataset, photo…)

GUIDE FEEDBACK / COMMENTS (record of the weekly meeting)

Action plan for next week

Task	Expected outcome	Owner	Due date

Guide Signature & Date	Team Lead Signature & Date

SEMESTER VIII · CP2 · WEEKLY PROGRESS LOG	
Week No.	03
Phase	Phase 1 — Re-baseline, Design Refinement & CDR
Dates (from – to)	
Planned agenda / focus for the week	

Member-wise work done this week (who did what)

Member (Name)	Student ID	Tasks undertaken this week	Prog.%
1.			
2.			
3.			

Actionable items completed

Challenges faced & how they were addressed

Evidence produced this week (repo/commit, drawing/schematic no., test-case ID, lab-notebook page, dataset, photo…)

GUIDE FEEDBACK / COMMENTS (record of the weekly meeting)

Action plan for next week

Task	Expected outcome	Owner	Due date

Guide Signature & Date	Team Lead Signature & Date

Phase 2 · Implementation of Remaining Objectives (O3–O5)	
Weeks	Weeks 4–8
Gate / Panel review	Continuous guide assessment
Accreditation coverage	PO4, PO5 · ABET SO6

2.1 Implementation progress

Task	Owner(s)	Start	End	% done	Dependencies

2.2 Deviations from design

Objective / module	Deviation	Reason	Impact

Technical Evidence — IT & Allied (Computing) (CP2 · objectives O3–O5)

Record the discipline-specific engineering evidence for the objectives built in this phase. Each item below has its own page(s); duplicate any table if you need more rows.

Requirements & Architecture (ISO/IEC/IEEE 29148 · 42010)

Req. ID	Type (F/NF)	Requirement / quality attribute	Priority	Traces to objective

Architecture / component & data-flow diagram

Note the ISO/IEC 25010 quality attributes (performance, security, reliability, maintainability…) the design addresses.

(Draw or paste the diagram here — include a title, labels and a legend.)

Data & Model Evidence (AI&DS / data projects)

Dataset(s)

Dataset / source	Size & split	Licence & provenance	Pre-processing / issues

Experiments

Experiment / run ID	Config / model	Metric	Baseline	Result vs target

Bias / fairness & data-privacy check (consent, anonymisation, DPDP/GDPR relevance)

Build & Version Control

Module / component	Owner	Repository / commit ref	CI-CD / build status

Security & Quality (OWASP / ISO 27001 · ISO/IEC 25010)

Control / quality attribute	Applicable? (Y/N)	How addressed / evidence

2.3 Unit test execution (O3–O5)

Test-case ID	Input	Expected output	Actual output	Pass/Fail

2.4 Issue tracker

Issue no.	Description	Impact	Owner	Status	Guide sign

Additional space — Implementation & testing (CP2)

Use this page to continue any table above or to record extra findings for this section. Note which item/row this continues.

SEMESTER VIII · CP2 · WEEKLY PROGRESS LOG	
Week No.	04
Phase	Phase 2 — Implementation of O3–O5
Dates (from – to)	
Planned agenda / focus for the week	

Member-wise work done this week (who did what)

Member (Name)	Student ID	Tasks undertaken this week	Prog.%
1.			
2.			
3.			

Actionable items completed

Challenges faced & how they were addressed

Evidence produced this week (repo/commit, drawing/schematic no., test-case ID, lab-notebook page, dataset, photo…)

GUIDE FEEDBACK / COMMENTS (record of the weekly meeting)

Action plan for next week

Task	Expected outcome	Owner	Due date

Guide Signature & Date	Team Lead Signature & Date

SEMESTER VIII · CP2 · WEEKLY PROGRESS LOG	
Week No.	05
Phase	Phase 2 — Implementation of O3–O5
Dates (from – to)	
Planned agenda / focus for the week	

Member-wise work done this week (who did what)

Member (Name)	Student ID	Tasks undertaken this week	Prog.%
1.			
2.			
3.			

Actionable items completed

Challenges faced & how they were addressed

Evidence produced this week (repo/commit, drawing/schematic no., test-case ID, lab-notebook page, dataset, photo…)

GUIDE FEEDBACK / COMMENTS (record of the weekly meeting)

Action plan for next week

Task	Expected outcome	Owner	Due date

Guide Signature & Date	Team Lead Signature & Date

SEMESTER VIII · CP2 · WEEKLY PROGRESS LOG	
Week No.	06
Phase	Phase 2 — Implementation of O3–O5
Dates (from – to)	
Planned agenda / focus for the week	

Member-wise work done this week (who did what)

Member (Name)	Student ID	Tasks undertaken this week	Prog.%
1.			
2.			
3.			

Actionable items completed

Challenges faced & how they were addressed

Evidence produced this week (repo/commit, drawing/schematic no., test-case ID, lab-notebook page, dataset, photo…)

GUIDE FEEDBACK / COMMENTS (record of the weekly meeting)

Action plan for next week

Task	Expected outcome	Owner	Due date

Guide Signature & Date	Team Lead Signature & Date

SEMESTER VIII · CP2 · WEEKLY PROGRESS LOG	
Week No.	07
Phase	Phase 2 — Implementation of O3–O5
Dates (from – to)	
Planned agenda / focus for the week	

Member-wise work done this week (who did what)

Member (Name)	Student ID	Tasks undertaken this week	Prog.%
1.			
2.			
3.			

Actionable items completed

Challenges faced & how they were addressed

Evidence produced this week (repo/commit, drawing/schematic no., test-case ID, lab-notebook page, dataset, photo…)

GUIDE FEEDBACK / COMMENTS (record of the weekly meeting)

Action plan for next week

Task	Expected outcome	Owner	Due date

Guide Signature & Date	Team Lead Signature & Date

SEMESTER VIII · CP2 · WEEKLY PROGRESS LOG	
Week No.	08
Phase	Phase 2 — Implementation of O3–O5
Dates (from – to)	
Planned agenda / focus for the week	

Member-wise work done this week (who did what)

Member (Name)	Student ID	Tasks undertaken this week	Prog.%
1.			
2.			
3.			

Actionable items completed

Challenges faced & how they were addressed

Evidence produced this week (repo/commit, drawing/schematic no., test-case ID, lab-notebook page, dataset, photo…)

GUIDE FEEDBACK / COMMENTS (record of the weekly meeting)

Action plan for next week

Task	Expected outcome	Owner	Due date

Guide Signature & Date	Team Lead Signature & Date

Phase 3 · Integration & System-Level Testing	
Weeks	Weeks 9–11
Gate / Panel review	Continuous guide assessment
Accreditation coverage	PO4, PO5 · ABET SO6

3.1 System integration (all objectives together)

Subsystems involved	Integration objective	Result	Issues identified

3.2 System-level test plan & execution

Test ID	Description (functional / non-functional / performance)	Expected	Actual	Pass/Fail

3.3 Experiment / investigation record (generate & analyse data)

Experiment	Variables / setup	Data / result	Analysis & conclusion

3.4 Standards-compliance validation

Standard	Validation method	Outcome	Remarks

Additional space — Integration & system testing

Use this page to continue any table above or to record extra findings for this section. Note which item/row this continues.

SEMESTER VIII · CP2 · WEEKLY PROGRESS LOG	
Week No.	09
Phase	Phase 3 — Integration & System-Level Testing
Dates (from – to)	
Planned agenda / focus for the week	

Member-wise work done this week (who did what)

Member (Name)	Student ID	Tasks undertaken this week	Prog.%
1.			
2.			
3.			

Actionable items completed

Challenges faced & how they were addressed

Evidence produced this week (repo/commit, drawing/schematic no., test-case ID, lab-notebook page, dataset, photo…)

GUIDE FEEDBACK / COMMENTS (record of the weekly meeting)

Action plan for next week

Task	Expected outcome	Owner	Due date

Guide Signature & Date	Team Lead Signature & Date

SEMESTER VIII · CP2 · WEEKLY PROGRESS LOG	
Week No.	10
Phase	Phase 3 — Integration & System-Level Testing
Dates (from – to)	
Planned agenda / focus for the week	

Member-wise work done this week (who did what)

Member (Name)	Student ID	Tasks undertaken this week	Prog.%
1.			
2.			
3.			

Actionable items completed

Challenges faced & how they were addressed

Evidence produced this week (repo/commit, drawing/schematic no., test-case ID, lab-notebook page, dataset, photo…)

GUIDE FEEDBACK / COMMENTS (record of the weekly meeting)

Action plan for next week

Task	Expected outcome	Owner	Due date

Guide Signature & Date	Team Lead Signature & Date

SEMESTER VIII · CP2 · WEEKLY PROGRESS LOG	
Week No.	11
Phase	Phase 3 — Integration & System-Level Testing
Dates (from – to)	
Planned agenda / focus for the week	

Member-wise work done this week (who did what)

Member (Name)	Student ID	Tasks undertaken this week	Prog.%
1.			
2.			
3.			

Actionable items completed

Challenges faced & how they were addressed

Evidence produced this week (repo/commit, drawing/schematic no., test-case ID, lab-notebook page, dataset, photo…)

GUIDE FEEDBACK / COMMENTS (record of the weekly meeting)

Action plan for next week

Task	Expected outcome	Owner	Due date

Guide Signature & Date	Team Lead Signature & Date

Phase 4 · Results & Impact Evaluation	
Weeks	Weeks 12–13
Gate / Panel review	Continuous guide assessment
Accreditation coverage	PO3, PO6, PO10 · ABET SO2, SO4

4.1 Outcome — planned vs achieved (per objective)

ID	Planned success criterion	Achieved result + evidence	Variance & reason	Status
O1				
O2				
O3				
O4				
O5				

4.2 Results analysis & benchmarking

How do the achieved results compare against the targets and against existing/benchmark solutions?

4.3 Impact — intended vs created

SDG target & intended impact	What was actually demonstrated	Evidence (pilot / user test / simulation / stakeholder)	Gap to full impact + pathway

Be honest: a 2-semester project can usually demonstrate an outcome, but rarely full real-world impact. State clearly what was proven and what would be needed to achieve the full intended impact.

SEMESTER VIII · CP2 · WEEKLY PROGRESS LOG	
Week No.	12
Phase	Phase 4 — Results & Impact Evaluation
Dates (from – to)	
Planned agenda / focus for the week	

Member-wise work done this week (who did what)

Member (Name)	Student ID	Tasks undertaken this week	Prog.%
1.			
2.			
3.			

Actionable items completed

Challenges faced & how they were addressed

Evidence produced this week (repo/commit, drawing/schematic no., test-case ID, lab-notebook page, dataset, photo…)

GUIDE FEEDBACK / COMMENTS (record of the weekly meeting)

Action plan for next week

Task	Expected outcome	Owner	Due date

Guide Signature & Date	Team Lead Signature & Date

SEMESTER VIII · CP2 · WEEKLY PROGRESS LOG	
Week No.	13
Phase	Phase 4 — Results & Impact Evaluation
Dates (from – to)	
Planned agenda / focus for the week	

Member-wise work done this week (who did what)

Member (Name)	Student ID	Tasks undertaken this week	Prog.%
1.			
2.			
3.			

Actionable items completed

Challenges faced & how they were addressed

Evidence produced this week (repo/commit, drawing/schematic no., test-case ID, lab-notebook page, dataset, photo…)

GUIDE FEEDBACK / COMMENTS (record of the weekly meeting)

Action plan for next week

Task	Expected outcome	Owner	Due date

Guide Signature & Date	Team Lead Signature & Date

Phase 5 · Dissemination, Documentation & Final Defense	
Weeks	Weeks 14–15
Gate / Panel review	Final Defense @ Week 15
Accreditation coverage	PO6, PO7, PO9, PO10 · ABET SO2, SO3, SO4

5.1 Dissemination

Output	Reference / link	Status

e.g., code repository, working demo/video, poster, conference/journal paper, patent, product page.

5.2 Individual contribution log (guide-verified)

Sl	Member	Objectives owned & key deliverables	Approx %	Guide verified
1				
2				
3				

5.3 Final ethics / data / IP confirmation

Confirm final data handling, licensing, consent and integrity; note any changes since CP1

SEMESTER VIII · CP2 · WEEKLY PROGRESS LOG	
Week No.	14
Phase	Phase 5 — Dissemination, Documentation & Defense
Dates (from – to)	
Planned agenda / focus for the week	

Member-wise work done this week (who did what)

Member (Name)	Student ID	Tasks undertaken this week	Prog.%
1.			
2.			
3.			

Actionable items completed

Challenges faced & how they were addressed

Evidence produced this week (repo/commit, drawing/schematic no., test-case ID, lab-notebook page, dataset, photo…)

GUIDE FEEDBACK / COMMENTS (record of the weekly meeting)

Action plan for next week

Task	Expected outcome	Owner	Due date

Guide Signature & Date	Team Lead Signature & Date

SEMESTER VIII · CP2 · WEEKLY PROGRESS LOG	
Week No.	15
Phase	Phase 5 — Dissemination, Documentation & Defense
Dates (from – to)	
Planned agenda / focus for the week	

Member-wise work done this week (who did what)

Member (Name)	Student ID	Tasks undertaken this week	Prog.%
1.			
2.			
3.			

Actionable items completed

Challenges faced & how they were addressed

Evidence produced this week (repo/commit, drawing/schematic no., test-case ID, lab-notebook page, dataset, photo…)

GUIDE FEEDBACK / COMMENTS (record of the weekly meeting)

Action plan for next week

Task	Expected outcome	Owner	Due date

Guide Signature & Date	Team Lead Signature & Date

Capstone Project 2 — Guide Evaluation Rubric

Continuous (guide) + review (panel) assessment. Guide circles the band and enters a score out of 10 for each criterion; the weighted total gives the phase grade.

Performance band	What it means (applies to every criterion)
Exemplary (9–10)	Complete, rigorous and well-evidenced; independent, professional-quality work that exceeds expectations.
Proficient (7–8)	Meets all expectations with sound method and adequate evidence; minor gaps only.
Developing (5–6)	Partially meets expectations; method or evidence incomplete; needs guidance to progress.
Inadequate (<5)	Missing, superficial or unsupported; does not meet the expected standard.

Criterion	Wt %	Evidence to look for	Band / Score /10
Design refinement & CDR	10	Justified changes; sound detailed design of O3–O5; design frozen	
Implementation of O3–O5	20	Remaining objectives built and unit-tested with evidence	
Integration & system-level testing	20	Full system integrated and validated at system level	
Results: outcome planned vs achieved	15	Honest, evidenced results against success criteria	
Impact & SDG contribution	10	Demonstrated impact + honest gap analysis (overclaiming penalised)	
Standards, ethics & individual contribution	10	Standards validated; ethics final; balanced, verified contribution	
Dissemination, documentation & defense	15	Outputs produced; complete diary; defense performance	
Total	**100**		

Overall guide comments & areas for improvement

Guide Signature & Date	

CP2 Reflection & Future Work

Critical reflection on the whole capstone journey — technical, teamwork and personal learning

Future work — what a next team should do to take this further

Section 4 · Standards, Outcome Mapping & Evidence Index

1.1 Programme Outcome (PO) mapping

Mapped under NBA GAPC v4.0 (Revised SAR 2025) — 11 Programme Outcomes, Washington Accord 2021 aligned. Institutions transitioning from 12 POs: old PO6+PO7 merge into new PO6; old PO8–PO12 become new PO7–PO11.

PO	Programme Outcome	Where addressed in this project (evidence)	Level (1-3)
PO1	Engineering Knowledge		
PO2	Problem Analysis		
PO3	Design/Development of Solutions		
PO4	Conduct Investigations of Complex Problems		
PO5	Engineering Tool Usage		
PO6	The Engineer and the World (incl. sustainability)		
PO7	Ethics (incl. diversity & inclusion)		
PO8	Individual and Collaborative Team Work		
PO9	Communication		
PO10	Project Management and Finance		
PO11	Life-Long Learning		

1.2 Programme-Specific Outcome (PSO) mapping — this cluster

PSO	Programme-Specific Outcome	Where addressed in this project (evidence)	Level (1-3)
PSO1			
PSO2			
PSO3			

1.3 Standards register (project-wide)

Standard / code	Clause / version	Where applied	Compliance evidence

1.4 Evidence index

List every evidence item referenced in this diary so an examiner can verify any claim.

Evidence ID	Description	Type	Location / link	Objective

Annex · Cluster A — Programme-Specific Notes

Applicable programmes
B.Tech CSE · CS&IT · AI&DS · IoT (software track)

Indicative standards & codes for this cluster

- ISO/IEC/IEEE 12207 — Software life-cycle processes
- ISO/IEC/IEEE 29148 — Requirements engineering (SRS)
- ISO/IEC/IEEE 42010 — Architecture description
- ISO/IEC/IEEE 29119 — Software testing
- ISO/IEC 25010 — Software product quality
- ISO/IEC 27001 / OWASP — Information security
- ACM / IEEE-CS Software Engineering Code of Ethics
- GDPR / India DPDP Act 2023 — Data privacy

Add your programme's specific standards and codes above; the guide will confirm the applicable set at Gate G0.